# Final Project - Part 1 of 3

35 Points Possible

4/30/2026

Attempt 1

In Progress

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**Unlimited Attempts Allowed**

**Details**

# COE 311K Final Project - Part 1 of 3

# Numerical Solution of Second-Order ODEs

## Overview

In this project, you will apply numerical methods to solve a real-world engineering problem modeled by a second-order ordinary differential equation (ODE). You will implement two numerical solution methods (Euler's Forward and RK4), analyze their performance, and conduct a stability analysis.

## Special Notes on the Final Project

This can be either a group project or an individual project. But we require only one submission. In the header section of your notebook, list all group members - name and EID. If this is an individual project, you will be given 3 bonus points. However, academic integrity rules apply.

## Learning Objectives

By completing this project, you will:

- Connect theoretical ODE concepts to real-world engineering applications
- Implement and compare numerical solution methods
- Analyze stability and accuracy of numerical schemes
- Develop technical communication skills through a structured Jupyter Notebook

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## Project Requirements

### 1. Model Selection

Choose **ONE** of the following real-world applications:

Option A: Spring-Mass System (Mechanical Oscillations)

$$m\frac{d^2y}{dt^2} + c\frac{dy}{dt} + ky = 0$$

Where:

- $m$ = mass (kg)
- $c$ = damping coefficient (N·s/m)
- $k$ = spring constant (N/m)
- $y(t)$ = displacement from equilibrium (m)

**Example applications:** Car suspension systems, building vibrations, mechanical oscillators

### Option B: RLC Electrical Circuit

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = V(t)$$

Where:

- $L$ = inductance (H)
- $R$ = resistance (Ω)
- $C$ = capacitance (F)
- $q(t)$ = charge on capacitor (C)
- $V(t)$ = external voltage (V)

**Example applications:** Audio filters, signal processing circuits, oscillators

### Option C: Damped Mechanical System

$$\frac{d^2y}{dt^2} + 2\zeta\omega\frac{dy}{dt} + \omega^2y = 0$$

Where:

- $\zeta$ = damping ratio (dimensionless)
- $\omega$ = natural frequency (rad/s)
- $y(t)$ = displacement or response

**Example applications:** Earthquake response of structures, shock absorbers, door closers

### Option D: Population Dynamics

$$\frac{d^2P}{dt^2} = rP - kP^2$$

Where:

- $P(t)$ = population size
- $r$ = growth rate

- $k$ = carrying capacity factor

**Example applications:** Predator-prey systems, ecological modeling, resource management

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# 2. Parameter Research

You must:

- **Research realistic parameter values** from engineering handbooks, scientific literature, or manufacturer specifications, you may use AI as well, but verify and cite where the AI got its information from
- **Cite your sources** for all parameter values
- **Justify your choices** based on the specific real-world scenario you're modeling
- **Define appropriate initial conditions** consistent with your scenario

**Example:** If modeling a car suspension, specify the make/model, cite the vehicle mass, research typical spring constants for that vehicle class, and choose initial conditions representing hitting a speed bump.

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# 3. Numerical Implementation

## Convert to First-Order System

Your second-order ODE must be converted to a system of two first-order ODEs:

For example, if $\frac{d^2y}{dt^2} = f(t, y, \frac{dy}{dt})$, let:

- $y_1 = y$
- $y_2 = \frac{dy}{dt}$

Then:

$$\frac{dy_1}{dt} = y_2$$

$$\frac{dy_2}{dt} = f(t, y_1, y_2)$$

## Implement Two Methods

**Euler's Forward Method:**

```
For each time step:
    y_1(t + h) = y_1(t) + h * y_2(t)
    y_2(t + h) = y_2(t) + h * f(t, y_1(t), y_2(t))
```

**Fourth-Order Runge-Kutta (RK4):**

```
For each time step, calculate four slopes (k1, k2, k3, k4)
and combine them with weights: (k1 + 2k2 + 2k3 + k4)/6
```

Requirements:

- Implement both methods in separate, well-documented functions
- Test with **at least three different step sizes** (e.g., h = 0.1, 0.05, 0.01)
- Simulate for a physically meaningful time interval

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## 4. Stability Analysis

Your analysis must include:

### a) Step Size Investigation

- Demonstrate how solution accuracy changes with step size
- Identify the **maximum stable step size** for each method
- Show what happens when the step size is **too large** (unstable behavior)

### b) Error Quantification

- If an analytical solution exists, calculate and plot the error
- Create a **log-log plot** of error vs. step size to verify convergence rates:
  - Euler's method: $O(h) \rightarrow$ slope $\approx 1$
  - RK4: $O(h^4) \rightarrow$ slope $\approx 4$
- If no analytical solution exists, use the finest step size solution as a reference

### c) Physical Interpretation

- Discuss whether numerical instabilities lead to **non-physical results**
  - For conservative systems: Does total energy increase artificially?
  - For damped systems: Does the solution grow instead of decay?
- Relate stability to the **physical behavior** of your system

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# Deliverable: Jupyter Notebook

Submit a **single Jupyter Notebook to your GitHub**(.ipynb file) **and submit a PDF to Canvas** with the following structure:

## Section 1: Introduction & Model Selection (~0.5 pages)

- Brief description of your chosen real-world system
- The governing second-order ODE
- Conversion to a system of first-order ODEs
- Physical meaning of each variable and parameter

## Section 2: Parameter Research & Justification (~0.5 pages)

- Table of all parameters with values, units, and sources

- Initial conditions with justification
- Description of the physical scenario being modeled

## Section 3: Numerical Methods Implementation (~0.25 pages + code)

- Well-commented Python functions for:
  - Euler's Forward method
  - RK4 method
- Brief explanation of your implementation approach

## Section 4: Solutions & Comparison (~0.5 pages + plots)

- Results from both methods with multiple step sizes
- Comparison plots (position vs. time, velocity vs. time)
- Phase portraits (if applicable)
- Discussion of visual differences between methods

## Section 5: Stability Analysis (~0.5 pages + plots)

- Demonstration of stability limits for each method
- Error analysis with appropriate plots
- Discussion of stable vs. unstable behavior
- Physical interpretation of results

## Section 6: Conclusions (~0.25 pages)

- Which method performed better for your problem?
- Computational cost vs. accuracy tradeoff
- Practical recommendations
- Lessons learned

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# Formatting Guidelines

## Length

- **Total length:** 5-7 pages when printed (including code and plots)
- **Text/markdown:** ~2 pages
- **Code:** ~1-2 pages
- **Plots:** ~2-3 pages (4-6 key figures)

## Code Requirements

- Use clear variable names
- Include comments explaining key steps
- Use modular functions (avoid copy-paste code)
- Follow Python best practices

## Plot Requirements

- All plots must have:
  - Descriptive titles
  - Labeled axes with units
  - Legends (when multiple curves are shown)
  - Readable font sizes
- Use `matplotlib` or similar plotting library

## Citations

- Use any standard citation format (IEEE, APA, or inline links)

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# Grading Rubric (35 points)

## Model & Parameters (10 points)

- Appropriate model selection (4 pts)
- Well-researched, realistic parameters (4 pts)
- Proper citations and justification (2 pts)

## Code Implementation (10 points)

- Correct Euler's Forward method implementation (4 pts)
- Correct RK4 implementation (4 pts)
- Code quality: comments, modularity, style (2 pts)

## Stability Analysis (10 points)

- Correct step size experiments (4 pts)
- Clear demonstration of stability limits (4 pts)
- Meaningful error analysis with appropriate plots (2 pts)

## Format (5 points)

- Clear, well-labeled plots with captions (2 pts)
- Concise, well-written explanations (2 pts)
- Overall organization and flow (1 pts)

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# Submission Instructions

- **Due Date:** 04/30/2026
- **Submit via GitHub:** Upload your .ipynb file
- **Submit via Canvas:** Upload your .pdf of your completed notebook
- **File naming:** `LastName_FirstName_COE311K_FinalProject.ipynb/LastName_FirstName_COE311K_FinalProject.pdf`

- **If this is a group project:** We only need one submission per group. In the header comments, identify each member (name and EID)
- **Ensure your notebook runs completely:** Before submitting, restart the kernel and run all cells to verify there are no errors

**Choose a submission type**

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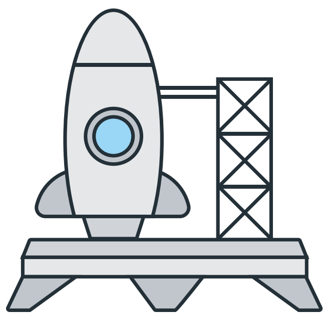

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